

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2023-P2-Model)** **Date: 22-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-26** **Max. Marks: 180**

22-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2023-P2)_GTA-26_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2023-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. Consider the function defined implicitly by the equation $y^2-2ye^{\sin^{-1}x}+x^2-1+[x]+e^{2\sin^{-1}x}=0,\quad -1<x<1$, Area of the region bounded by the curve is (Where [x] denotes the greatest integer function)

A) $\pi+1 sq.units$ **B)** $\pi-1 sq.units$ **C)** $\frac{\pi}{2}+1 sq.units$ **D)** $\frac{\pi}{2}-1 sq.units$

2. The solution of the differential equation $x^2y\frac{d^2y}{dx^2}+\left(x\frac{dy}{dx}-y\right)^2=0$ is

A) $y=\sqrt{x(c_2x+2c_1)}$ **B)** $y=\sqrt{x(c_1-2c_2x)}$ **C)** $y=\pm\sqrt{x(c_2x-2c_1)}$ **D)** $y=\pm\sqrt{x(\frac{c_2}{x}-2c_1)}$

3. k black balls are placed in a row with gaps between them. The balls are then numbered such that n^{th} ball from the left has (n^2+n) imprinted on it. Now white balls are placed in the gaps such that the number of white balls in a gap is equal to the difference of the numbers on the two consecutive black balls between which the gap occurs. If number of white balls available is 340, then number of black balls is

A) 13 **B)** 17 **C)** 18 **D)** 19

4. Let $5x-3y=8\sqrt{2}$ be the normal at the point $P\left(\frac{5}{\sqrt{2}},\frac{3}{\sqrt{2}}\right)$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$. If M, M' are the feet of the perpendiculars from the foci S, S' respectively on the tangent at P, then the point of intersection of SM' and S'M is _____.

A) $\left(\frac{5}{2},0\right)$ **B)** $\left(0,\frac{5}{2}\right)$ **C)** $\left(\frac{41}{10\sqrt{2}},\frac{3}{2\sqrt{2}}\right)$ **D)** $\left(\frac{3}{2\sqrt{2}},\frac{41}{10\sqrt{2}}\right)$

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Let f(x) be a non-positive continuous function and $F(x)=\int_0^x f(t)dt\ \forall x\geq 0$ and $f(x)\geq cF(x)$ where $c>0$ and let $g:[0,\infty)\to\mathbb{R}$ be a function such that $\frac{d(g(x))}{dx}<g(x)\ \forall x>0$ and g(0) = 0, then

A) The total number of roots of the equation $f(x)=g(x)$ is 1

B) The total number of roots of the equation $f(x)=g(x)$ is 0

C) The number of solution of equation $|x^2+x-6|=f(x)+g(x)$ is 1

D) The number of solution of equation $|x^2+x-6|=f(x)+g(x)$ is 0

6. Let matrix $A=\begin{bmatrix} x & y & -z \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$, where $x,y,z\in\mathbb{N}$. If $|\text{adj}(\text{adj}(\text{adj}(\text{adj}A)))|=4^8\cdot 5^{16}$, then the number of such matrix is p and value of $\int_{-\pi/2}^{\pi/2}2^{\sin x}dx+\int_{5/2}^{4}\sin^{-1}(\log_2(x-2))dx$ is equal to $\frac{q\pi}{4}$. *If* $\vec{a}=\hat{i}+\hat{j}+2\hat{k},\vec{b}=\hat{i}+2\hat{j}+2\hat{k}$ *and* $|\vec{c}|=1$ such that $[\vec{a}\times\vec{b}\quad\vec{b}\times\vec{c}\quad\vec{c}\times\vec{a}]$ has maximum value then

A) The value of $|(\vec{a}\times\vec{b})\times\vec{c}|^2$ *is* 0

B) $p=36$

C) $q=5$

D) $p-q=20$

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7. Let f be a cubic polynomial function such that

$$\begin{bmatrix} a^2 & 4a & 1 \\ b^2 & 4b & 1 \\ c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} f(-1) & 0 \\ f(1) & f(0) \\ f(2) & 1 \end{bmatrix}=\begin{bmatrix} 2a^2+16a+17 & 4a+1 \\ 2b^2+16b+17 & 4b+1 \\ 2c^2+16c+17 & 4c+1 \end{bmatrix}, \text{ then}$$

A) Number of real roots of f(x) = 0 is 1

B) The sum of greatest and least values of $f(x)$ *in* $[-1,1]$ is 5

C) $y=f(x)$ is a bijective function

D) There are two values of k for which the equation f(x) = k has exactly two real and distinct roots

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

8. If k is the sum of all values of x such that $\cos^3 3x+\cos^3 5x=8\cos^3 4x\cos^3 x$ where x is measured in degrees and $100<x<200$, then $\frac{1}{3}\times(\text{sum of digits of } k)$ equals

9. If the equation in x, $\left(\cos^{-1}\frac{x}{2}\right)^2+\pi\sin^{-1}\left(\frac{x}{2}\right)+\frac{\pi^2}{12}(x^2+6x+8)-\left(\sin^{-1}\frac{x}{2}\right)^2=\frac{\pi^2}{4}(2k-1)$ possesses solution, then the number of integral values of k is

10. OA, OB, OC are the sides of a rectangular parallelepiped whose diagonals are OO', AA', BB', CC'. D is the centre of rectangle $AC'O'B'$ *and* D' is the centre of rectangle $O'A'CB'$. If sides OA, OB, OC are in the ratio of 1:2:3 and $\angle DOD'=\alpha$, then $\frac{697}{72}\cos^2\alpha$ is: (Centre of rectangle is midpoint of diagonal)

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11. Three biased coins $C_1, C_2 and\ C_3$ lie on a table. Their respective probabilities of falling heads when tossed once are 1/3, 2/3 and 1. A coin is picked up randomly, tossed twice and observed to fall heads on both the occasions. Now given that a coin has been tossed twice and observed to fall heads both times. If the probability that a 3rd throw of the same coin will also produce heads is $\frac{a}{b}$ where $a\ and\ b$ are relatively prime, find the value of (a^2+b^2).

12. Let $a_1, a_2, a_3 \ldots a_n\ (n \in \mathbb{N})$ is a sequence such that $a_1 = a_2 = 2$ and $\frac{2a_{n-1}a_n}{a_{n-1}a_{n+1} - a_n^2} = n^2 - n,\ \forall n = 2,3,4,\ldots,$ find the value of $\frac{[S]}{13}$, where $S = \sum_{k=2}^{20} \frac{a_k}{3a_k - a_{k+1}}$ and $([\cdot] denotes\ G.I.F.)$

13. Consider, $f(x) = (|2x-1| + |x| + |2x+1|)\left[\frac{1}{4}(x^4 - 2x^2 + 4)\right]$ where $[k]$ denotes the greatest integer less than or equal to k. If k_1 be the number of points where $f(x)$ is discontinuous in $\left[\frac{-3}{2}, \frac{3}{2}\right]$ k_2 be the number of points where $f(x)$ is non-derivable in $\left(\frac{-3}{2}, \frac{3}{2}\right)$ then $k_1 + k_2$ is

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Let $f: \left(\frac{-\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$, such that f(x) is differentiable at $x = 0$ and $f(x) = \begin{cases} \frac{1}{2}(1+\sec x) f\left(\frac{x}{2}\right), & x \neq 0 \\ 2, & x = 0 \end{cases}$

14. The value of $\pi\left(f\left(\frac{\pi}{4}\right) + f\left(-\frac{\pi}{4}\right)\right)$ is

15. The value of $\lim_{x \to 0}\left(\left\{\frac{f(x)-2}{x^2}\right\} + \left[\frac{f(x)-2\cos x}{x^2}\right]\right)$ is

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PARAGRAPH-II

Consider two planes $P_1: 2x-3y+z=-5$ and $P_2: x-4y-14z=5$. If a line L perpendicular to P_1 whose one point $A(-7, y_1, z_1)$ lies on plane P_1 and point $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ lies on plane P_2 as well as on line L.

16. Let equation of line L be $\frac{x+7}{a}=\frac{y-y_1}{-b}=\frac{z-z_1}{c}$ where $a,b,c \in \mathbb{N}$ then least value of $a+b+c$ is

17. Value of $\frac{z_3-z_2}{y_3-y_2}+\frac{x_3-x_2}{z_3-z_2}$ is

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PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

18. A dose containing radioactive ^{18}F is scheduled to be prepared at 6.10 AM and 1.00 mL of it is to be given to a patient at 8.00 AM. If the patient reaches clinic late and takes the dose at 8.55 AM, how much quantity of the dose should be given to the patient? Half-life of ^{18}F is 110 min.

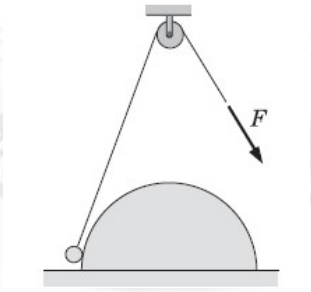
A) 1.41 mL **B)** 2.00 mL **C)** 2.82 mL **D)** None of these

19. When an α – particle collides with a stationary lithium nucleus, the following nuclear reaction may take place $^{4}\text{He} + ^{7}\text{Li} \rightarrow ^{10}\text{B} + \text{n}$

If the Q-value of this reaction is $E = -2.87$ MeV, find the minimum value of kinetic energy of incoming α – particles required for the reaction.

A) 3.51 MeV **B)** 4.51 MeV **C)** 5.51 MeV **D)** 6.51 MeV

20. A small metal ball is being pulled gradually on a fixed frictionless hemisphere as shown in the figure. Radii of the ball and that of the pulley are much smaller than that of the hemisphere. As the ball slides from the bottom to a position close to the top of the hemisphere, how do the magnitudes of pulling force F and contact force R between the ball and the hemisphere change?



A) F increases and R decreases. **B)** F decreases and R increases.

C) F decreases and R remains unchanged. **D)** F remains unchanged and R decreases.

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21. Identical balloons are connected through identical valves with two identical cylinders. One cylinder contains gaseous helium and the other gaseous nitrogen. Both the gases may be assumed ideal, and both the cylinders weigh equally. Which balloon will be inflated faster when the valves are opened, and why?

A) The nitrogen balloon will be inflated faster because nitrogen is a heavier element, and so the molecules of nitrogen have greater momentum and thus force the balloon to expand at a greater rate.

B) The helium balloon will be inflated faster, because helium is a lighter element and so its atoms move faster and can get into the balloon at a greater rate.

C) The helium balloon will be inflated faster because the helium must be at higher pressure and hence the gas will be forced into the balloon at a greater rate.

D) It will depend on whether the gases have to flow up or down to enter the balloon. Helium being lighter than air as compared to nitrogen will rise faster than nitrogen. Therefore, if the balloons are at the top of the cylinders, the helium balloon will be inflated faster; and if they are at the bottoms, the nitrogen balloon will be inflated faster.

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

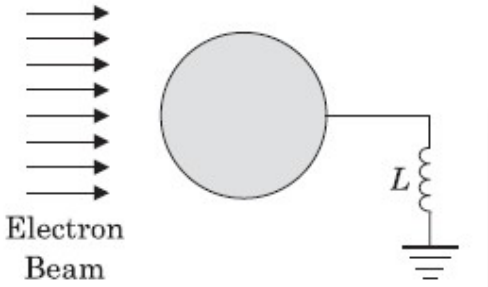
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
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22. An uncharged conducting ball of radius r is earthed through an inductor L as shown in the figure. A horizontal electron beam swoops on the ball from a great distance. The density of electrons in the incident beam is n and all these electrons are moving with velocity v that is much smaller than the speed of light so the relativistic effects can be neglected. Which of the following statements is/are correct?

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A) Maximum charge on the ball is equal to $2ne\pi r^2 v\sqrt{\pi\varepsilon_0 rL}$

B) Maximum charge on the ball is equal to $4ne\pi r^2 v\sqrt{\pi\varepsilon_0 rL}$

C) Maximum current through the inductor is equal to $2ne\pi r^2 v$

D) Maximum current through the inductor is equal to $4ne\pi r^2 v$

23. An isolated zinc bead of radius r charged to a negative potential V_0 (assuming potential at infinitely distant points to be zero) placed in free space is being continuously irradiated by ultraviolet light of wavelength λ. Photoelectric threshold wavelength for zinc is λ_0, speed of light is c, Denoting Planck's constant by h, permittivity of free space by ε_0, charge and mass of an electron by $-e$ and m, select correct statements from the following

A) Maximum speed of a photoelectron immediately after emission is $\sqrt{\frac{2hc}{m}\left(\frac{1}{\lambda}-\frac{1}{\lambda_0}\right)}$.

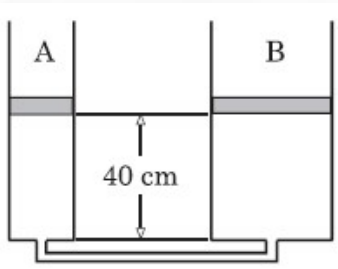
B) Maximum speed of a photoelectron at a great distance from the bead is $\sqrt{\frac{2hc}{m}\left(\frac{1}{\lambda}-\frac{1}{\lambda_0}\right)+\frac{2eV_0}{m}}$

C) Potential of the bead after prolonged irradiation is $-\frac{hc}{e}\left(\frac{1}{\lambda}-\frac{1}{\lambda_0}\right)$

D) Total number of photoelectrons emitted from the bead is $\frac{4\pi\varepsilon_0 r}{e}\left\{\frac{hc}{e}\left(\frac{1}{\lambda}-\frac{1}{\lambda_0}\right)+V_0\right\}$

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24. Two vertical conducting cylinders A and B of different cross sections are connected by a thin tube as shown. An ideal gas is trapped in the cylinders by two airtight pistons of masses 1.0 kg and 2.0 kg respectively. Initially the pistons are at the same height 40 cm above the base of the cylinders. If an additional mass of 10 g were gently placed on the piston in the cylinder A, which of the following statements correctly describe the new steady state ultimately reached.



A) Pressure of the gas has been increased.
B) Pressure of the gas remains unchanged.
C) Height difference between the pistons becomes 30 cm.
D) Height difference between the pistons becomes 60 cm.

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

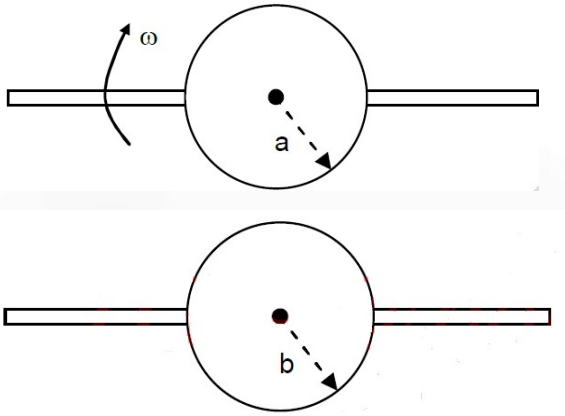
Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

25. A tuning fork of frequency 200 Hz is in unison with a sonometer wire. Then the number of beats heard in 30 s if the tension is increased by 1% is $10x$. Then x is

26. Two uniform spheres shown in given figure rotate separately on two parallel horizontal axis passing though their centres. The upper sphere has an angular speed, and the lower sphere is at rest. Now the two spheres are moved together so that their surfaces touch. After a short time the two spheres, now in contact, are rotating without slipping. Find the final rate of rotation of upper sphere in rad/s [given moment of inertia of the upper sphere $I_1 = 4\ kg\,m^2$, moment of inertia of lower sphere $I_2 = 2kg\ m^2$ radius of upper sphere a = 2 m, radius of lower sphere b = 1 m, $\omega_0 = 12$ rad/s].

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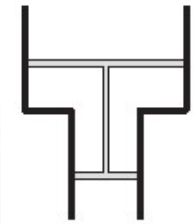


27. A glass of refractive index 1.5 is coated with a thin layer of thickness t, of refractive index 1.8. Light of wavelength λ travelling in air is incident normally on the layer. It is partly reflected at the upper and at the lower surfaces of the layer. The two reflected rays interfere. If λ = 648 nm, obtained the least value of t as $x\left(10^{-8}\right)\text{m}$ for which the rays interfere constructively. Then x is

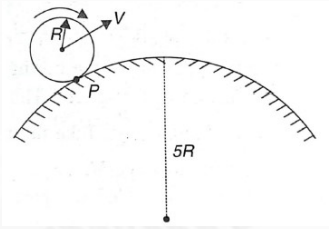
28. An ideal mono-atomic gas is confined in a cylinder with the help of a piston that can slide inside the cylinder without friction. Walls of the cylinder have finite conductivity. Initially the piston is in equilibrium with its external surface exposed to the atmosphere at pressure 10^5 Pa. A transient pulse of heat is injected into the cylinder through its walls causing the gas to expand until its volume changes by 0.2 L. Then the heat supplied by the pulse was $(10x)J$ then x is

29. A vertical tube made of highly conducting material has two portions of cross-sectional area A_1 and $A_2\,(A_2 > A_1)$. Two pistons connected by a rod of length ℓ can slide in these portions without friction. Total mass of the pistons and the rod is m. Between the pistons, n moles of an ideal gas is trapped. Atmospheric pressure is p_0, acceleration of free fall is g and universal gas constant is R. If the piston-rod assembly is shifted slightly from its equilibrium position and then released, it starts oscillating. Then the angular frequency of these oscillations is $\frac{xmg + yp_0\left(A_2 - A_1\right)}{\sqrt{mnRT}}$ where T is temperature of gas. Then $x+y$ is

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30. A convex surface has a uniform radius of curvature equal to 5R. A wheel of radius R is rolling without sliding on it with a constant speed V. Then the acceleration of the point (P) of the wheel which is in contact with the convex surface is $x(2.5)ms^{-2}$. Then x is



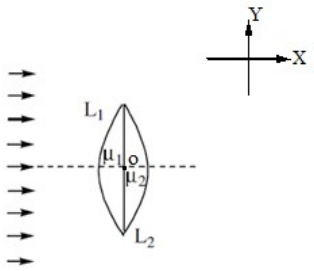
(V = 6 meter/second, R = 4 meter)

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Two identical Plano-convex lenses $L_1(\mu_1 = 1.4)$ and $L_2(\mu_2 = 1.5)$ of radii of curvature $R = 20$ cm are placed as shown in figure.



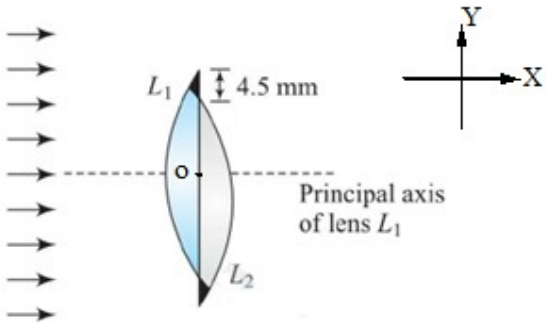
O is Origin of coordinate system

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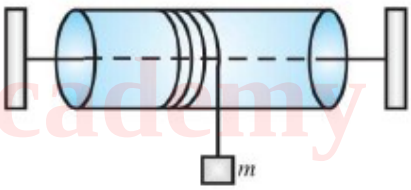
31. Then the position (X- coordinate) of the image of the parallel beam of light relative to the common principal axis (in cm) is

32. Now, the second lens is shifted vertically downward (from origin O) by a small distance 4.5 mm and the extended parts of L_1 *and* L_2 are blackened as shown in figure. Then the magnitude of Y coordinate of new position of the image of the parallel beam (in mm) is .



PARAGRAPH-II

A cord is wound round the circumference of a solid cylinder of radius R and mass M. The axis of the cylinder is horizontal. A weight mg is attached to the end of the cord and falls from rest. After falling through a distance h.



33. The angular velocity of the cylinder will be $\omega = \sqrt{\frac{xmgh}{(M+ym)R^2}}$ then xy is

34. If the mass starts from rest and falls a distance h, then its speed v at that instant is $v = \sqrt{\frac{xmgh}{(M+ym)}}$ then $x+y$ is

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CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

35. Consider the species: $O_2, O_2^+, O_2^-, O_2^{2-}, N_2, NO, CO, CN^-$.

Using molecular orbital theory, identify the correct statement(s):

A) The bond order in NO is 2.5, and the molecule is paramagnetic due to one unpaired electron in an antibonding π^* MO.

B) CN^- and N_2 are isoelectronic, both have bond order 2, and CN^- has HOMO as a σ bonding MO.

C) O_2^{2-} has the highest bond order among O_2, O_2^-, O_2^{2-} and is diamagnetic.

D) In CO, the HOMO is a π bonding orbital largely localised on carbon, which explains $CO's$ reactivity as a π donor ligand.

36. Consider the following statements related to colloids.

(I) Lyophobic colloids are **not** formed by simple mixing of dispersed phase and dispersion medium.

(II) For emulsions, both the dispersed phase and the dispersion medium are liquid.

(III) Micelles are produced by dissolving a surfactant in any solvent at any temperature.

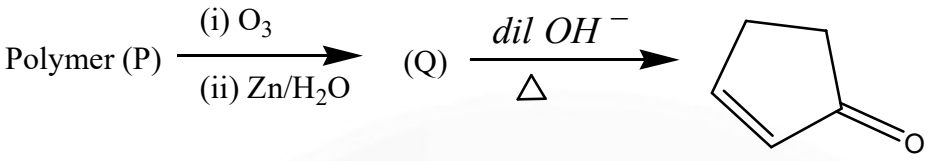
(IV) Tyndall effect can be observed from a colloidal solution with dispersed phase having the same refractive index as that of the dispersion medium.

The option with the correct set of statements is

A) (I) and (II) **B)** (II) and (III) **C)** (III) and (IV) **D)** (II) and (IV)

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37.

Incorrect for above reaction sequence is :

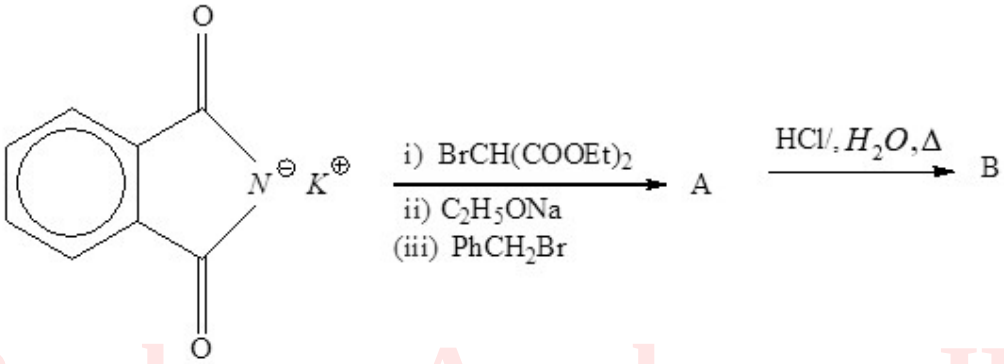
(A) (P) exist in both cis and trans forms

(B) (P) is a 1,4-addition polymer of 2-methylbuta-1,3-diene

(C) Q can show both Tollens's and Iodoform test.

(D) Q gives positive Iodoform test but no reaction with $NaHSO_3$

38. Consider the following sequence of reaction,



The major final product (B) is :

A) Essential aminoacid with double bond Equivalent of '4'

B) Non-essential aminoacid with double bond Equivalent of '4'

C) Essential aminoacid with double bond Equivalent of '5'

D) Non-essential aminoacid with double bond Equivalent of '5'

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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39. KCl crystalizes similar to rock salt where k^+ ions are at the body centre and edge centre .

A) The nearest neighbour of Cl^- ion is $6\,K^+$ ion

B) The packing fraction is given by $\frac{1.732\pi}{8}$

C) The nearest neighbors of K^+ ions are $6\,Cl^-$ ions

D) The second nearest neighbours of K^+ ions are $12\,K^+$ ions.

40. Which of the following statements about the **types of isomerism** exhibited by following complexes are correct?

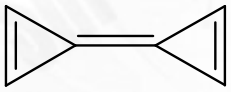
A) $[Cr(en)_2(NCS)(NO_2)]^+$ exhibits geometrical isomerism, optical isomerism, and linkage isomerism.

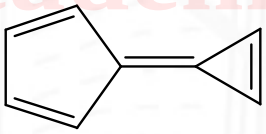
B) HCl/, H_2O represents an *ionisation isomeric pair*. Treatment of each isomer with excess $AgNO_3$ gives a same mass of $AgBr$ precipitate per mole of complex.

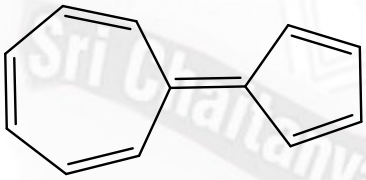
C) $[Pt(py)(NH_3)(NO_2)(Cl)]$ has exactly **3** geometric isomers $(cis, cis\,\&\,trans)$ since it is square planar with 4 different ligands.

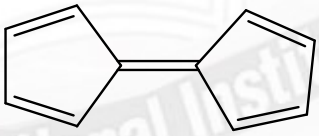
D) $\left[Ir(PPh_3)_2\,Cl(CO)\right]$ can exhibit optical isomerism

41. Which of the following compound(s) can exhibit C-C free rotation?

A) 

B) 

C) 

D) 

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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42. H_2S (5 moles) reacts completely with acidified aqueous potassium permanganate solution. In this reaction, the number of moles of water produced is **x**, and the number of moles of electrons involved is **y**.

The value of $(x+y)$ is ____.

43. Among the following species, the total number having an sp^3 hybridised central atom is___.

$[I_3]^+, [SiO_4]^{4-}, SO_2Cl_2, XeF_2, SF_4, ClF_3, Ni(CO)_4, XeO_2F_2, [PtCl_4]^{2-}, XeF_4,$

$SOCl_2, [ICl_4]^-, OSF_2, [BH_4]^-, IF_5, [XeF_5]^-, POCl_3, [IBr_2]^-, SF_3^+, ClO_4^-$

44. Consider the following species/molecules:

$Br_3O_8, F_2O, H_2S_4O_6, H_2S_5O_6, C_3O_2, N_2H_4, H_3PO_3, K_2MnO_4,$
$Hg_2Cl_2, Fe_3O_4, Pb_3O_4, S_2Cl_2$

For each molecule/ion, count the number of atoms existing in **zero oxidation state**. Find the **sum** of all such atoms across all species.

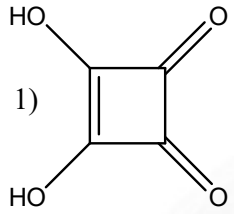
45. A plant virus is found to consist of uniform cylindrical particles of 150 Å in diameter and $5000\ Å$ long. The specific volume of the virus is $0.75\ cm^3/g$. If the virus is considered to be a single particle, The molar mass of virus is $A \times 10^B$ gram/mole, the value of A + B (answer in 2 significant figures) is

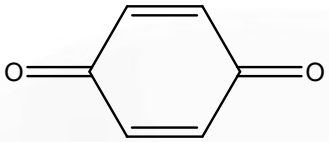
46. 50 mL *of* 0.2 molal urea solution $(density = 1.012\ g\ mL^{-1}\ at\ 300\ K)$ is mixed with 250 mL of a solution containing 0.06 g of urea. Both the solutions were prepared in the same solvent. The osmotic pressure (in Torr) of the resulting solution at 300 K is ___. (answer in 3 significant figures)

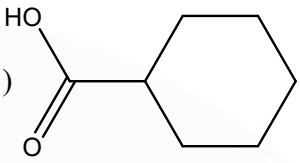
[Use: Molar mass of urea $= 60\ g\ mol^{-1}$; gas constant, $R = 62\ L\ Torr\ K^{-1}\ mol^{-1}$; Assume, $\Delta_{mix}H = 0, \Delta_{mix}V = 0$]

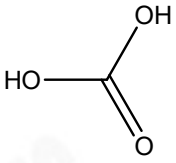
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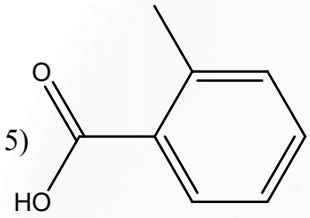
47. Compounds having less P^{Ka} value than phenol.

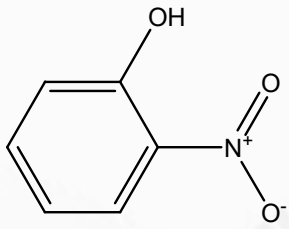
1) 

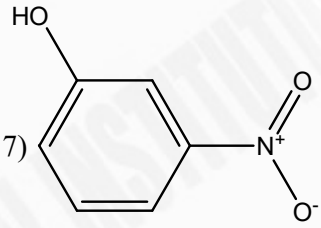
2) 

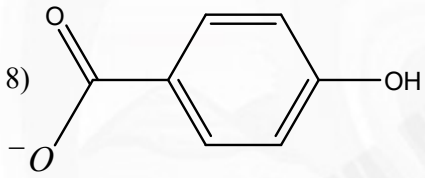
3) 

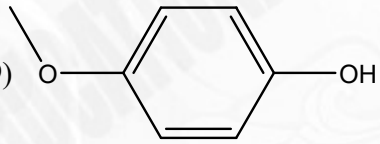
4) 

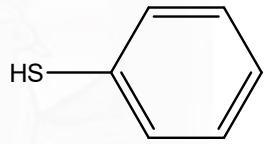
5) 

6) 

7) 

8) 

9) 

10) 

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

A trinitro compound, 1,3,5-tris-(4-nitrophenyl) benzene, on complete reaction with an excess of Sn/HCl gives a major product, which on treatment with an excess of $NaNO_2$/HCl *at* $0^{\circ}C$ provides **P** as the product. **P**, upon treatment with excess of H_2O at room temperature, gives the product **Q**. Bromination of **Q** in aqueous medium furnishes the product **R**. The compound **P** upon treatment with an excess of phenol under basic conditions gives the product **S**.

The molar mass difference between compounds **Q** and **R** is $474\ g\ mol^{-1}$ and between compounds **P** and **S** is $172.5\ g\ mol^{-1}$.

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48. The number of heteroatoms present in one molecule of **R** is _____.

Use: Molar mass (in g mol^{-1}): H = 1, C = 12, N = 14, O = 16, Br = 80, Cl = 35.5

Atoms other than C *and* H are considered as heteroatoms

49. The total number of carbon atoms and heteroatoms present in one molecule of **S** is _____. Use: Molar mass (in g mol^{-1}): H = 1, C = 12, N = 14, O = 16, Br = 80, Cl = 35.5

(Atoms other than C *and* H are considered as heteroatoms)

PARAGRAPH-II

2 mol *of* $Hg(g)$ is combusted in a fixed volume bomb calorimeter with excess of O_2 at 298 K *and* 1 atm into $HgO(s)$. During the reaction, temperature increases from 298.0 K *to* 312.8 K. If heat capacity of the bomb calorimeter and enthalpy of formation of $Hg(g)$ *are* 20.00 kJ K^{-1} *and* 61.32 kJ mol^{-1} *at* 298 K, respectively. ($R=8.3\ J\ mol^{-1}\ K^{-1}$)

50. The magnitude of standard molar enthalpy of formation of $HgO(s)$ *at* 298 K *in* kJ/mol is

51. The magnitude of $|\Delta U|$ *in kJ for* $2Hg(g)+O_2(g)\rightarrow 2HgO(s)$ in Bomb calorimeter is

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30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048248*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	745	NUMBER OF QUALIFIED RANKS	4,212

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